

Chapter 1.6 Expected Value

Rick Durrett Probability: Theory and Examples

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Built from the local Chapter 1 LLMviki flash cards.

Roadmap for 1.6

Learning goal

Treat expectation as integration and deploy inequalities, change of variables, and limit theorems confidently.

1. Expectation as Lebesgue integral.
2. Linearity, monotonicity, and convergence theorems.
3. Markov, Chebyshev, Jensen, and moment inequalities.
4. Change of variables, tail expectations, inclusion-exclusion, and Bonferroni.

Expected Value as Integral

Definition

For a random variable X on $(\Omega, \mathcal{F}, \mathbb{P})$,

$$\mathbb{E}X = \int_{\Omega} X d\mathbb{P},$$

provided the positive and negative parts do not both have infinite integral.

Pitfall

$\mathbb{E}X$ is not defined when $\mathbb{E}X^+ = \mathbb{E}X^- = \infty$.

LLMwiki: Expected value as Lebesgue integral; ID: durrett_ch1_expectation_as_integral

Linearity and Monotonicity

Theorem

If X, Y are integrable and $a, b \in \mathbb{R}$, then

$$\mathbb{E}(aX + bY) = a\mathbb{E}X + b\mathbb{E}Y.$$

If $X \leq Y$ a.s., then $\mathbb{E}X \leq \mathbb{E}Y$ whenever the expectations are defined.

This is exactly integral linearity and monotonicity applied on the probability space. Independence is not needed for linearity; independence enters only when factoring products.

LLMwiki: [Linearity and monotonicity of expectation](#); ID: `durrett_ch1_expectation_linearity`

Markov and Chebyshev

Markov

If $Y \geq 0$ and $a > 0$, then

$$\mathbb{P}(Y \geq a) \leq \frac{\mathbb{E}Y}{a}.$$

Chebyshev

If $\mathbb{E}X = \mu$ and $\text{var}(X) = \sigma^2$, then

$$\mathbb{P}(|X - \mu| \geq a) \leq \frac{\sigma^2}{a^2}.$$

LLMwiki: Markov and Chebyshev inequalities; ID: durrett_ch1_markov_chebyshev

Markov: How to Prove

For $Y \geq 0$,

$$a\mathbf{1}_{\{Y \geq a\}} \leq Y.$$

Taking expectations gives $a\mathbb{P}(Y \geq a) \leq \mathbb{E}Y$. Chebyshev follows by applying Markov to $Y = (X - \mu)^2$.

LLMwiki: Markov and Chebyshev inequalities; ID: durrett_ch1_markov_chebyshev

Expectation Limit Theorems

Toolkit

Fatou, monotone convergence, dominated convergence, and bounded convergence all transfer to expectations.

- Fatou: nonnegative sequences give lower semicontinuity.
- MCT: $0 \leq X_n \uparrow X$ implies $\mathbb{E}X_n \uparrow \mathbb{E}X$.
- DCT: $X_n \rightarrow X$ and $|X_n| \leq Y$ with $\mathbb{E}Y < \infty$ implies $\mathbb{E}X_n \rightarrow \mathbb{E}X$.

[LLMwiki: Limit theorems for expectation](#); ID: `durrett_ch1_expectation_limit_theorems`

Change of Variables Formula

Theorem

If X has law μ and f is measurable, then

$$\mathbb{E}f(X) = \int f(x) \mu(dx),$$

whenever $f \geq 0$ or $f(X)$ is integrable.

Prove first for indicators $f = \mathbf{1}_A$, where both sides equal $\mathbb{P}(X \in A)$. Extend by linearity to simple functions, then by MCT to nonnegative f , and by signed decomposition to integrable f .

LLMwiki: [Change of variables formula](#); ID: [durrett_ch1_change_of_variables](#)

Tail Integral Formula

Formula

For $X \geq 0$,

$$\mathbb{E}X = \int_0^\infty \mathbb{P}(X > t) dt.$$

Write

$$X(\omega) = \int_0^\infty \mathbf{1}_{\{t < X(\omega)\}} dt.$$

Then apply Tonelli to interchange the ω and t integrals.

LLMwiki: Tail integral and nonnegative expectations; ID: durrett_ch1_tail_sum_nonnegative

Inclusion-Exclusion

Formula

For events $A_1, \dots, A_n$,

$$\mathbb{P}\left(\bigcup_i A_i\right) = \sum_i \mathbb{P}(A_i) - \sum_{i < j} \mathbb{P}(A_i A_j) + \cdots + (-1)^{n-1} \mathbb{P}\left(\bigcap_i A_i\right).$$

Use

$$\mathbf{1}_{\bigcup_i A_i} = 1 - \prod_{i=1}^n (1 - \mathbf{1}_{A_i}),$$

expand the finite product, then take expectations.

LLMwiki: [Inclusion-exclusion and Bonferroni](#); ID: `durrett_ch1_inclusion_exclusion`

Exercise 1.6.1

Suppose ϕ is strictly convex. Under the assumptions of Jensen's inequality, show that

$$\phi(EX) = E\phi(X)$$

implies $X = EX$ almost surely.

LLMwiki: Equality case in strict Jensen; ID: `exercise_method_strict_jensen_equality`

Exercise 1.6.1: Proof Skeleton

Let $m = \mathbb{E}X$. If $\mathbb{P}(X \neq m) > 0$, then X is nonconstant on a set of positive probability. Strict convexity makes Jensen strict for nonconstant random variables:

$$\mathbb{E}\phi(X) > \phi(\mathbb{E}X).$$

This contradicts equality. Hence $\mathbb{P}(X = m) = 1$.

LLMwiki: Equality case in strict Jensen; ID: `exercise_method_strict_jensen_equality`

Exercise 1.6.2

Suppose $\phi : \mathbb{R}^n \rightarrow \mathbb{R}$ is convex. Show that

$$E\phi(X_1, \dots, X_n) \geq \phi(EX_1, \dots, EX_n),$$

provided $E|\phi(X_1, \dots, X_n)| < \infty$ and $E|X_i| < \infty$ for all i .

LLMwiki: Jensen's inequality for integrals; ID: durrett_ch1_jensen_integral

Exercise 1.6.2: Proof Skeleton

Let $m = (\mathbb{E}X_1, \dots, \mathbb{E}X_n)$. A convex function has a supporting hyperplane at m :

$$\phi(x) \geq a \cdot x + b, \quad \phi(m) = a \cdot m + b.$$

Taking expectations yields

$$\mathbb{E}\phi(X) \geq a \cdot \mathbb{E}X + b = \phi(m).$$

LLMwiki: Jensen's inequality for integrals; ID: durrett_ch1_jensen_integral

Exercise 1.6.3

Chebyshev's inequality is and is not sharp.

1. Show that if $0 < b \leq a$ are fixed, there is X with $EX^2 = b^2$ for which $P(|X| \geq a) = b^2/a^2$.
2. Show that if $0 < EX^2 < \infty$, then

$$\lim_{a \rightarrow \infty} \frac{a^2 P(|X| \geq a)}{EX^2} = 0.$$

LLMwiki: Two-point distributions as Chebyshev extremizers; ID: exercise_method_two_point_extremizers

Exercise 1.6.3: Proof Skeleton

For fixed a, b , take

$$\mathbb{P}(X = a) = \mathbb{P}(X = -a) = \frac{b^2}{2a^2}, \quad \mathbb{P}(X = 0) = 1 - \frac{b^2}{a^2}.$$

Then $\mathbb{E}X^2 = b^2$ and $\mathbb{P}(|X| \geq a) = b^2/a^2$. For the limit statement set $Y = X^2$. Then

$$a^2 \mathbb{P}(|X| \geq a) \leq \mathbb{E}(Y; Y \geq a^2).$$

Since $Y\mathbf{1}_{\{Y \geq a^2\}} \rightarrow 0$ and is dominated by integrable Y , DCT gives the limit.

LLMwiki: Two-point distributions as Chebyshev extremizers; ID: exercise_method_two_point_extremizers

Exercise 1.6.4

One-sided Chebyshev bound.

1. Let $a > b > 0$, $0 < p < 1$, and let X satisfy $P(X = a) = p$, $P(X = -b) = 1 - p$. Apply Chebyshev's inequality to $\phi(x) = (x + b)^2$ and conclude that if Y has the same mean and variance as X , then $P(Y \geq a) \leq p$, with equality when $Y = X$.
2. If $EY = 0$, $\text{var}(Y) = \sigma^2$, and $a > 0$, show

$$P(Y \geq a) \leq \frac{\sigma^2}{a^2 + \sigma^2},$$

and show equality is possible.

LLMwiki: Two-point distributions as Chebyshev extremizers; ID: exercise_method_two_point_extremizers

Exercise 1.6.4: Proof Skeleton

For the two-point X ,

$$\mathbb{E}(X + b)^2 = p(a + b)^2.$$

If Y has the same mean and variance, then $\mathbb{E}(Y + b)^2$ has the same value. On $\{Y \geq a\}$,

$$(Y + b)^2 \geq (a + b)^2,$$

so Markov gives $P(Y \geq a) \leq p$. For mean zero and variance σ^2 , choose the extremal two-point distribution with values a and $-\sigma^2/a$. Its mass at a is

$$\frac{\sigma^2}{a^2 + \sigma^2},$$

giving both the bound and equality case.

LLMwiki: Two-point distributions as Chebyshev extremizers; ID: exercise_method_two_point_extremizers

Exercise 1.6.5

Show that the following lower bounds do not exist:

1. If $\varepsilon > 0$,

$$\inf\{P(|X| > \varepsilon) : EX = 0, \text{var}(X) = 1\} = 0.$$

2. If $y \geq 1$ and $\sigma^2 \in (0, \infty)$,

$$\inf\{P(|X| > y) : EX = 1, \text{var}(X) = \sigma^2\} = 0.$$

LLMwiki: Two-point distributions as Chebyshev extremizers; ID: exercise_method_two_point_extremizers

Exercise 1.6.5: Proof Skeleton

Fix $\delta > 0$ and put most mass at the mean. For part (i), take

$$\mathbb{P}(X = 0) = 1 - \delta, \quad \mathbb{P}(X = \delta^{-1/2}) = \mathbb{P}(X = -\delta^{-1/2}) = \delta/2.$$

Then $\mathbb{E}X = 0$ and $\text{var}(X) = 1$, while the tail event has probability at most δ . For part (ii), use $X = 1 + \sigma Z$, where Z is the variable from part (i). Then $\mathbb{E}X = 1$, $\text{var}(X) = \sigma^2$, and the event $\{|X| > y\}$ is contained in the rare nonzero part for small δ .

LLMwiki: Two-point distributions as Chebyshev extremizers; ID: `exercise_method_two_point_extremizers`

Exercise 1.6.6

Let $Y \geq 0$ with $EY^2 < \infty$. Apply Cauchy–Schwarz to $Y\mathbf{1}_{\{Y>0\}}$ and conclude

$$P(Y > 0) \geq \frac{(EY)^2}{EY^2}.$$

LLMwiki: Cauchy-Schwarz lower bound for positive probability; ID: exercise_method_paley_zygmund_basic

Exercise 1.6.6: Proof Skeleton

Since $Y = Y\mathbf{1}_{\{Y>0\}}$,

$$\mathbb{E}Y = \mathbb{E}(Y\mathbf{1}_{\{Y>0\}}).$$

Cauchy–Schwarz gives

$$(\mathbb{E}Y)^2 \leq \mathbb{E}(Y^2)\mathbb{E}(\mathbf{1}_{\{Y>0\}}^2) = \mathbb{E}Y^2 \mathbb{P}(Y > 0).$$

Divide by $\mathbb{E}Y^2$ unless it is zero, in which case the claim is trivial.

LLMwiki: Cauchy–Schwarz lower bound for positive probability; ID: `exercise_method_paley_zygmund_basic`

Exercise 1.6.7

Let $\Omega = (0, 1)$ with Lebesgue measure, let $\alpha \in (1, 2)$, and set

$$X_n = n^\alpha \mathbf{1}_{(1/(n+1), 1/n)}.$$

Show $X_n \rightarrow 0$ almost surely and Theorem 1.6.8 applies with $h(x) = x$ and $g(x) = |x|^{2/\alpha}$, but the X_n are not dominated by an integrable random variable.

LLMwiki: [Limit theorems for expectation](#); ID: `durrett_ch1_expectation_limit_theorems`

Exercise 1.6.7: Proof Skeleton

The intervals $(1/(n+1), 1/n)$ are disjoint, so each ω belongs to at most one of them; hence $X_n(\omega) \rightarrow 0$. Also

$$\mathbb{E}g(X_n) = n^2 \left(\frac{1}{n} - \frac{1}{n+1} \right) = \frac{n}{n+1} \leq 1,$$

and $|h(x)|/g(x) = |x|^{1-2/\alpha} \rightarrow 0$. If $|X_n| \leq Y$ with $\mathbb{E}Y < \infty$, then on $(1/(n+1), 1/n)$, $Y \geq n^\alpha$, so

$$\mathbb{E}Y \geq \sum_{n=1}^N n^\alpha \left(\frac{1}{n} - \frac{1}{n+1} \right),$$

whose terms are comparable to $n^{\alpha-2}$; the series diverges since $\alpha \in (1, 2)$.

LLMwiki: [Limit theorems for expectation](#); ID: [durrett_ch1_expectation_limit_theorems](#)

Exercise 1.6.8

Suppose a probability measure μ has

$$\mu(A) = \int_A f(x) dx$$

for all Borel A . Show that for any g with $g \geq 0$ or $\int |g| d\mu < \infty$,

$$\int g(x) \mu(dx) = \int g(x) f(x) dx.$$

LLMwiki: Change of variables formula; ID: durrett_ch1_change_of_variables

Exercise 1.6.8: Proof Skeleton

First prove it for $g = \mathbf{1}_A$:

$$\int \mathbf{1}_A d\mu = \mu(A) = \int_A f(x) dx.$$

Extend by linearity to nonnegative simple functions. For $g \geq 0$, take simple $g_n \uparrow g$ and apply MCT on both sides. For signed integrable g , apply the nonnegative result to g^+ and g^- and subtract.

[LLMwiki: Change of variables formula](#); ID: durrett_ch1_change_of_variables

Exercise 1.6.9

Let $A_1, \dots, A_n$ be events and $A = \bigcup_{i=1}^n A_i$. Prove the inclusion–exclusion formula.

LLMwiki: Indicator expansion for inclusion-exclusion and Bonferroni; ID:
`exercise_method_inclusion_exclusion_indicators`

Exercise 1.6.9: Proof Skeleton

Pointwise,

$$\mathbf{1}_A = 1 - \prod_{i=1}^n (1 - \mathbf{1}_{A_i}).$$

Expanding the product gives

$$\mathbf{1}_A = \sum_i \mathbf{1}_{A_i} - \sum_{i < j} \mathbf{1}_{A_i \cap A_j} + \sum_{i < j < k} \mathbf{1}_{A_i \cap A_j \cap A_k} - \cdots.$$

Taking expectations yields the finite inclusion–exclusion identity.

LLMwiki: Indicator expansion for inclusion-exclusion and Bonferroni; ID:
`exercise_method_inclusion_exclusion_indicators`

Exercise 1.6.10

Prove the Bonferroni inequalities for $A = \bigcup_{i=1}^n A_i$: stopping inclusion–exclusion after an even number of sums gives a lower bound, and stopping after an odd number of sums gives an upper bound.

LLMwiki: Indicator expansion for inclusion-exclusion and Bonferroni; ID:

`exercise_method_inclusion_exclusion_indicators`

Exercise 1.6.10: Proof Skeleton

For each outcome let

$$N = \sum_i \mathbf{1}_{A_i}.$$

Then $\mathbf{1}_A = \mathbf{1}_{\{N \geq 1\}}$. The inclusion–exclusion partial sums become

$$S_r(N) = \sum_{k=1}^r (-1)^{k-1} \binom{N}{k}.$$

For $N = 0$ all terms vanish. For $N \geq 1$, alternating binomial partial sums lie alternately above and below 1. Thus odd truncations dominate $\mathbf{1}_A$, even truncations are below it. Take expectations.

LLMwiki: Indicator expansion for inclusion-exclusion and Bonferroni; ID:
`exercise_method_inclusion_exclusion_indicators`

Exercise 1.6.11

If $E|X|^k < \infty$, show that for $0 < j < k$,

$$E|X|^j < \infty, \quad E|X|^j \leq (E|X|^k)^{j/k}.$$

LLMwiki: Jensen's inequality for integrals; ID: durrett_ch1_jensen_integral

Exercise 1.6.11: Proof Skeleton

Let $Y = |X|^k$ and $r = j/k \in (0, 1)$. The map $t \mapsto t^r$ is concave, so Jensen gives

$$\mathbb{E}Y^r \leq (\mathbb{E}Y)^r.$$

Since $Y^r = |X|^j$,

$$\mathbb{E}|X|^j \leq (\mathbb{E}|X|^k)^{j/k} < \infty.$$

LLM[wiki](#): Jensen's inequality for integrals; ID: durrett_ch1_jensen_integral

Exercise 1.6.12

Apply Jensen's inequality with $\phi(x) = e^x$ and $P(X = \log y_m) = p(m)$ to prove

$$\sum_{m=1}^n p(m)y_m \geq \prod_{m=1}^n y_m^{p(m)}$$

when $\sum_m p(m) = 1$ and $p(m), y_m > 0$.

LLMwiki: [Jensen's inequality for integrals](#); ID: durrett_ch1_jensen_integral

Exercise 1.6.12: Proof Skeleton

Let $\mathbb{P}(X = \log y_m) = p(m)$. Jensen for $\phi(x) = e^x$ gives

$$\mathbb{E}e^X \geq e^{\mathbb{E}X}.$$

Now

$$\mathbb{E}e^X = \sum_{m=1}^n p(m)y_m, \quad e^{\mathbb{E}X} = \exp\left(\sum_m p(m)\log y_m\right) = \prod_m y_m^{p(m)}.$$

LLMwiki: Jensen's inequality for integrals; ID: durrett_ch1_jensen_integral

Exercise 1.6.13

If $EX_1^- < \infty$ and $X_n \uparrow X$, show that

$$EX_n \uparrow EX.$$

LLMwiki: Limit theorems for expectation; ID: durrett_ch1_expectation_limit_theorems

Exercise 1.6.13: Proof Skeleton

Let $Y_n = X_n - X_1$. Then $Y_n \geq 0$ and $Y_n \uparrow X - X_1$. By MCT,

$$\mathbb{E}Y_n \uparrow \mathbb{E}(X - X_1).$$

Since $\mathbb{E}X_1^- < \infty$, adding $\mathbb{E}X_1$ is legitimate in the extended sense:

$$\mathbb{E}X_n = \mathbb{E}X_1 + \mathbb{E}Y_n \uparrow \mathbb{E}X.$$

LLMwiki: [Limit theorems for expectation](#); ID: `durrett_ch1_expectation_limit_theorems`

Exercise 1.6.14

Let $X \geq 0$, but do not assume $E(1/X) < \infty$. Show

$$\lim_{y \rightarrow \infty} yE(1/X; X > y) = 0, \quad \lim_{y \downarrow 0} yE(1/X; X > y) = 0.$$

LLMwiki: Tail truncation for expectation limits; ID: `exercise_method_expectation_tail_truncation`

Exercise 1.6.14: Proof Skeleton

As $y \rightarrow \infty$,

$$0 \leq y\mathbb{E}(1/X; X > y) = \mathbb{E}(y/X; X > y) \leq \mathbb{P}(X > y) \rightarrow 0.$$

As $y \downarrow 0$, fix $\eta > 0$ and choose $\delta > 0$ such that $\mathbb{P}(0 < X \leq \delta) < \eta$. For $0 < y < \delta$,

$$y\mathbb{E}(1/X; X > y) \leq \mathbb{P}(y < X \leq \delta) + y/\delta \leq \eta + y/\delta.$$

Let $y \downarrow 0$, then $\eta \downarrow 0$.

LLMwiki: Tail truncation for expectation limits; ID: exercise_method_expectation_tail_truncation

Exercise 1.6.15

If $X_n \geq 0$, show that

$$E \left(\sum_{n=0}^{\infty} X_n \right) = \sum_{n=0}^{\infty} EX_n.$$

LLMwiki: Limit theorems for expectation; ID: durrett_ch1_expectation_limit_theorems

Exercise 1.6.15: Proof Skeleton

Let

$$S_N = \sum_{n=0}^N X_n.$$

Then $S_N \uparrow \sum_{n=0}^{\infty} X_n$. MCT gives

$$\mathbb{E} \sum_{n=0}^{\infty} X_n = \lim_{N \rightarrow \infty} \mathbb{E} S_N.$$

Finite linearity gives $\mathbb{E} S_N = \sum_{n=0}^N \mathbb{E} X_n$; let $N \rightarrow \infty$.

[LLMviki: Limit theorems for expectation; ID: durrett_ch1_expectation_limit_theorems](#)

Exercise 1.6.16

If X is integrable and A_n are disjoint sets with union A , show

$$\sum_{n=0}^{\infty} E(X; A_n) = E(X; A),$$

i.e. the sum converges absolutely and has the displayed value.

LLMwiki: [Linearity and monotonicity of expectation](#); ID: durrett_ch1_expectation_linearity

Exercise 1.6.16: Proof Skeleton

Since X is integrable,

$$\sum_n \mathbb{E}(|X|; A_n) = \mathbb{E}(|X|; A) \leq \mathbb{E}|X| < \infty,$$

so $\sum_n \mathbb{E}(X; A_n)$ converges absolutely. For finite unions,

$$\sum_{n=0}^N \mathbb{E}(X; A_n) = \mathbb{E}\left(X; \bigcup_{n=0}^N A_n\right).$$

As $N \rightarrow \infty$, $X\mathbf{1}_{\bigcup_{n=0}^N A_n} \rightarrow X\mathbf{1}_A$ and is dominated by $|X|$. DCT gives the limit.

LLMwiki: [Linearity and monotonicity of expectation](#); ID: durrett_ch1_expectation_linearity